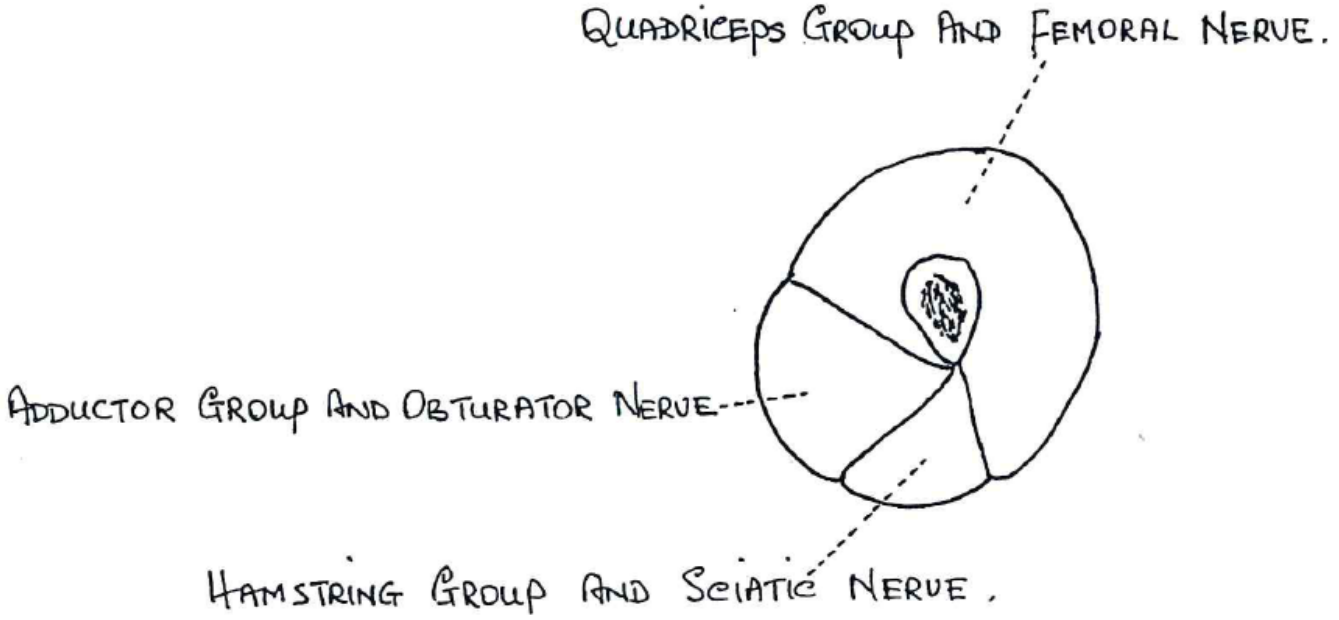
Posterior compartment of thigh

Learning Objectives

1. Enumerate the hamstring muscles of the posterior compartment of thigh
2. Appreciate the attachments and actions of hamstring muscles
3. Know the blood supply and innervation of posterior compartment of thigh
4. Interpret sonographic appearance of hamstring muscles
5. Know the relations of the sciatic nerve
6. List the boundaries, floor, roof and contents of the popliteal fossa
7. Produce a differential diagnosis of a popliteal swelling


QUADRICEPS GROUP AND FEMORAL NERVE.
ADDUCTOR GROUP AND OBTURATOR NERVE
HAMSTRING GROUP AND SCIATIC NERVE.

Muscles

- Biceps femoris
- Semimembranosus
- Semitendinosus
- Ischial part of adductor magnus

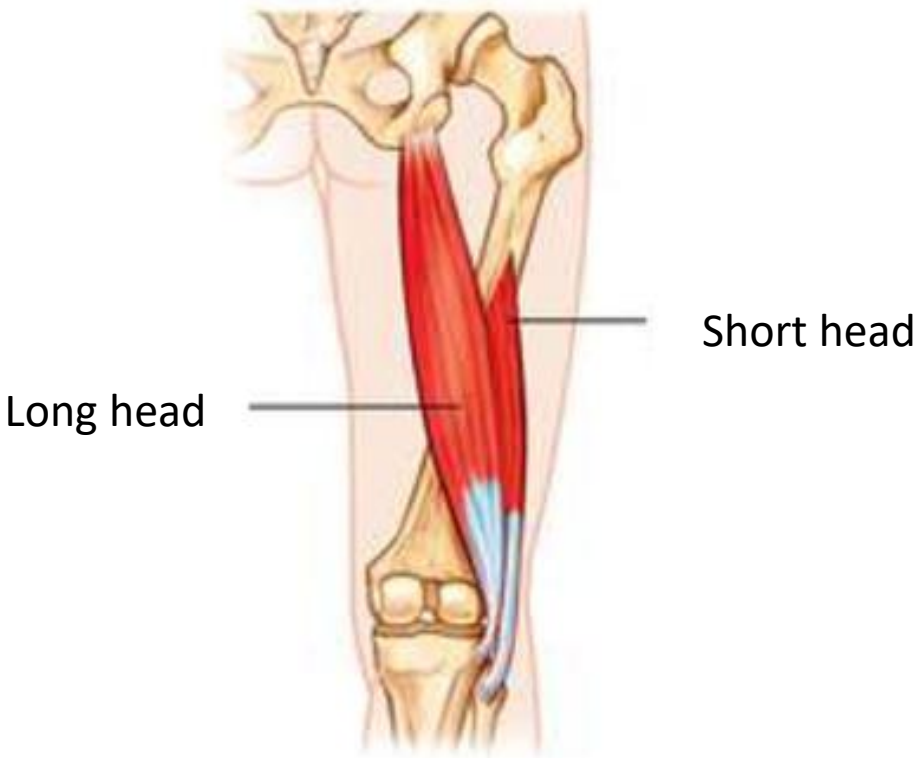
Blood supply

Branches of profunda femoris

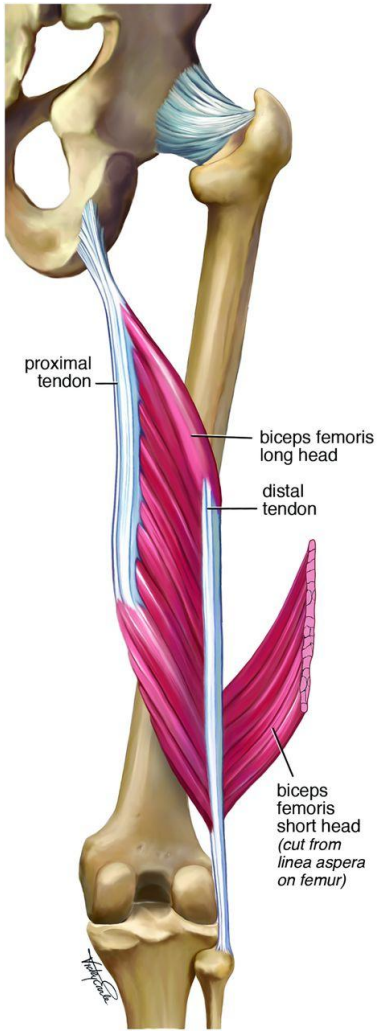
Nerve supply

Sciatic Nerve

Biceps femoris



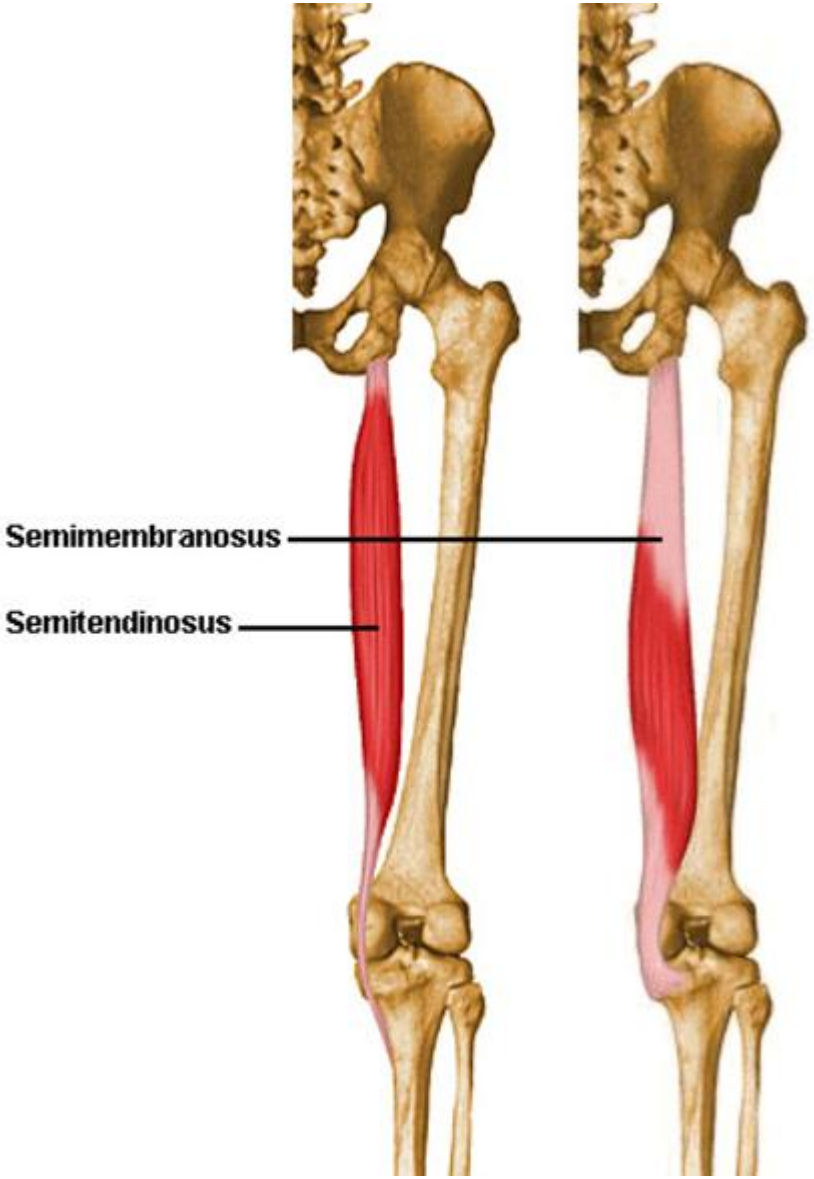
Proximal and distal tendons extending into the muscle belly of biceps femoris.

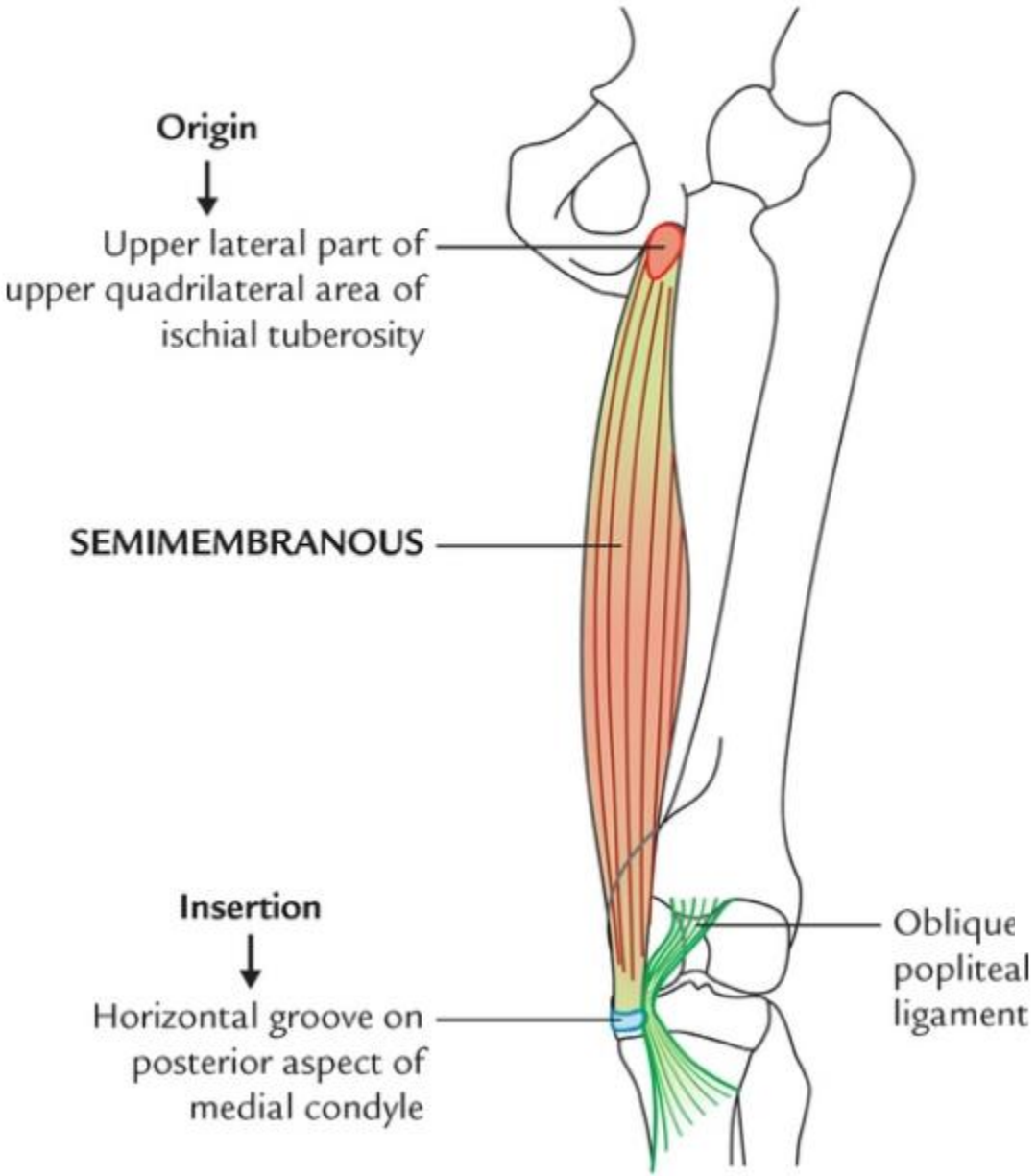


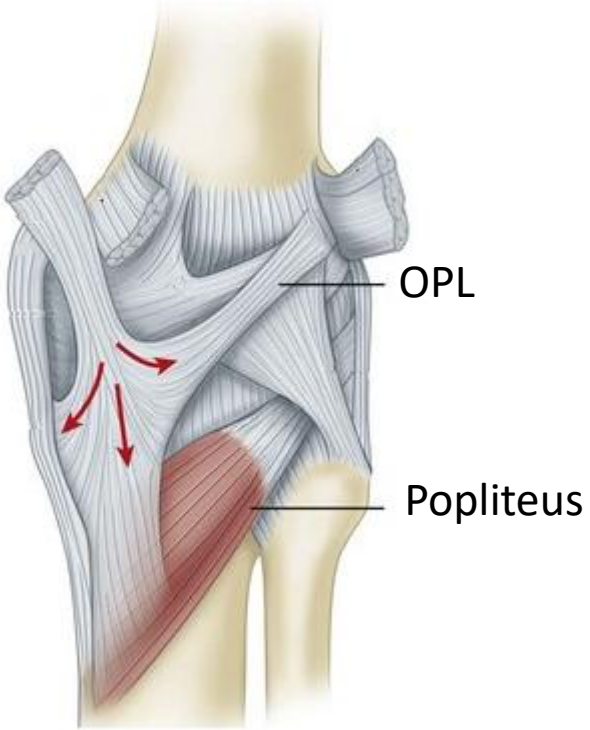
Peter Brukner, and David Connell Br J Sports Med 2016;50:205-208

BJSM

Semimembranosus and Semitendinosus

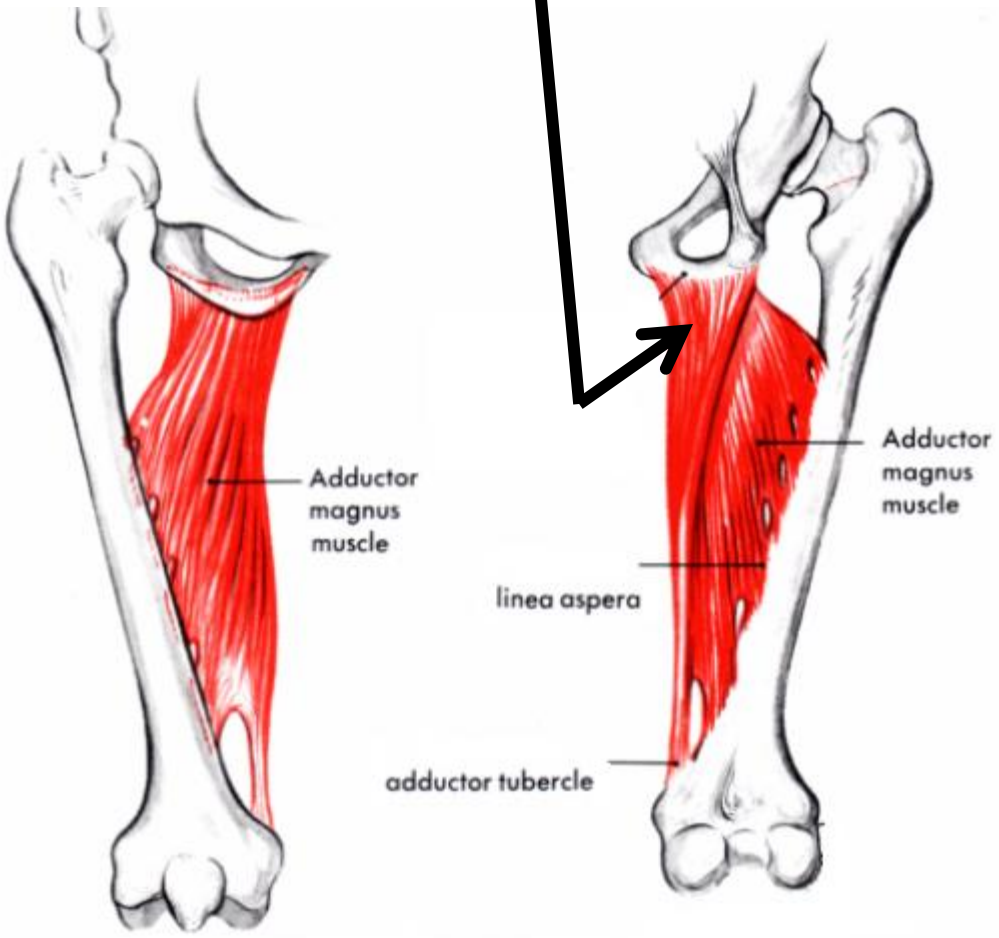






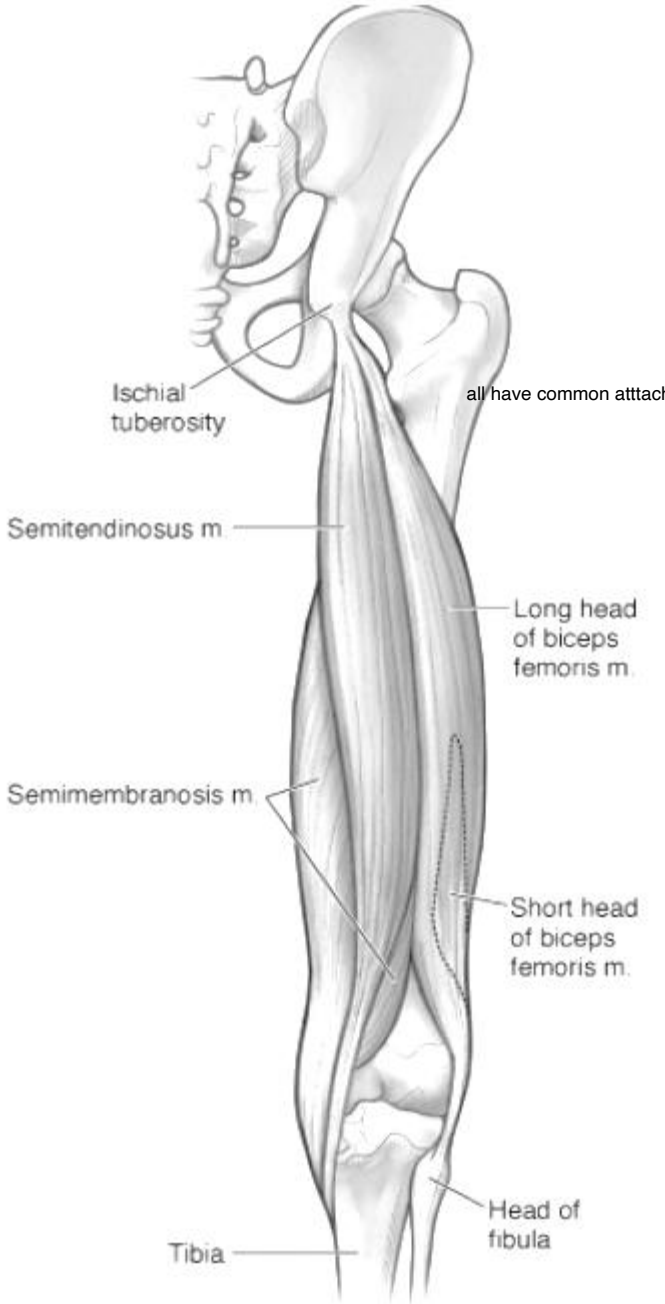
Musculoskeletal Key
Fastest Musculoskeletal Insight Engine

Adductor magnus ischial part



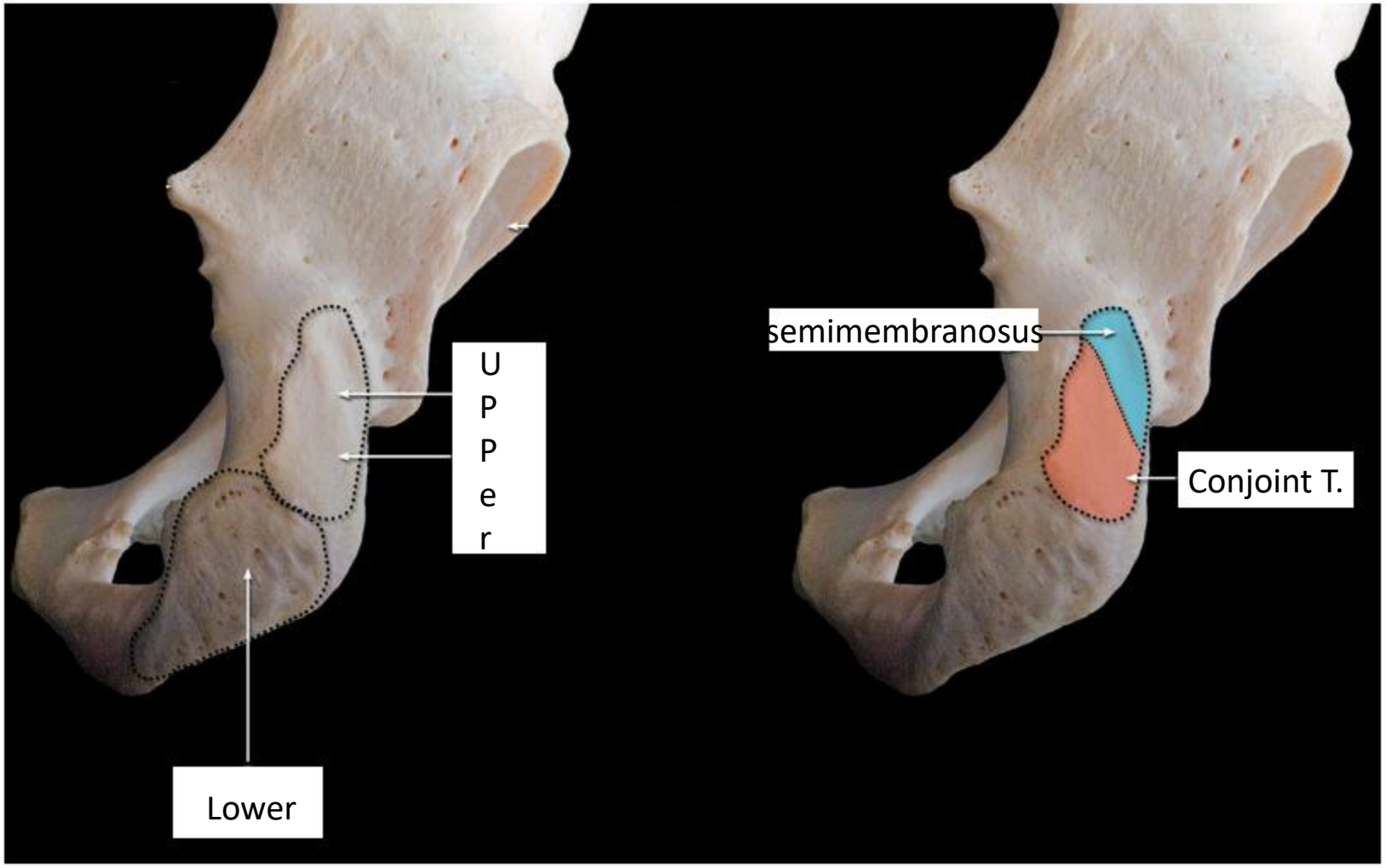
anterior

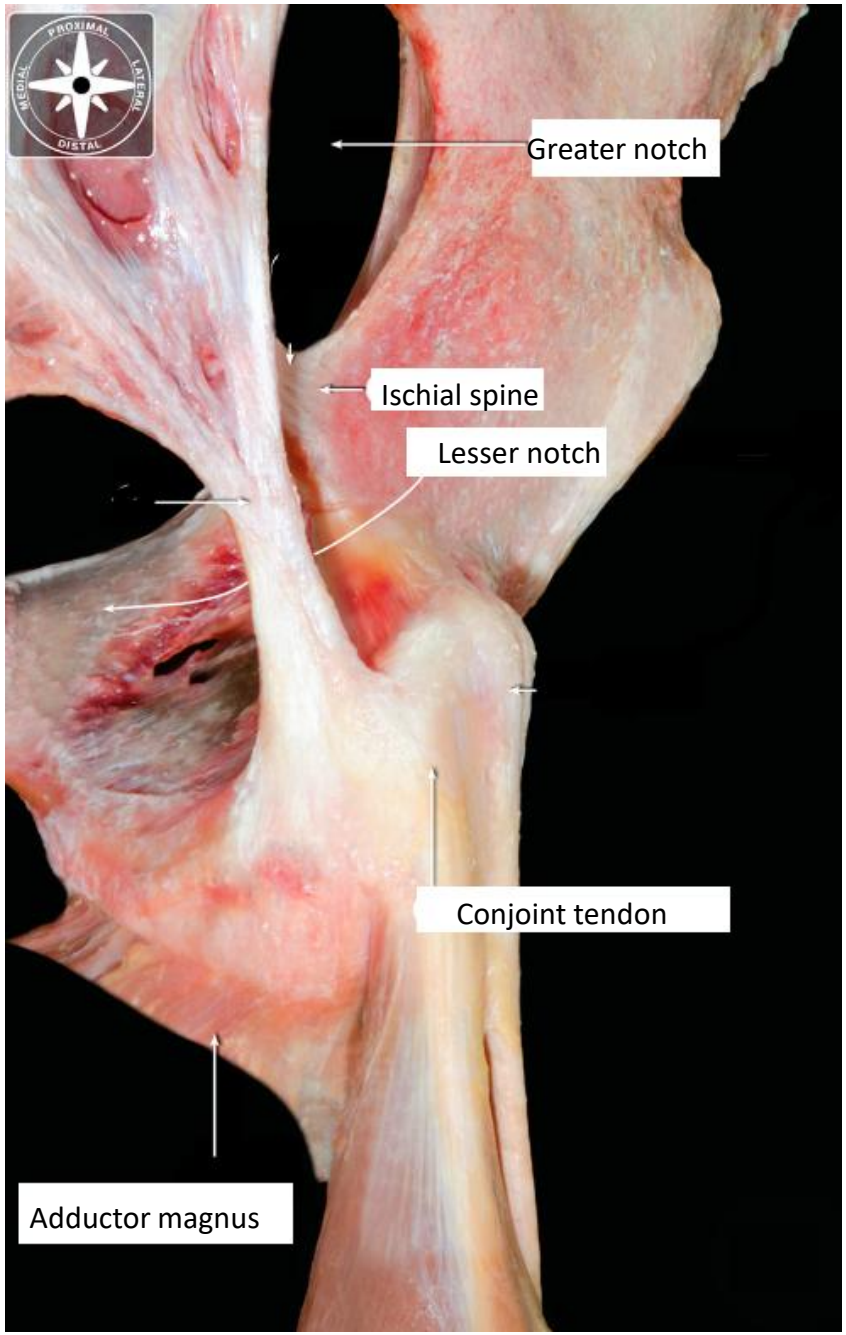
posterior

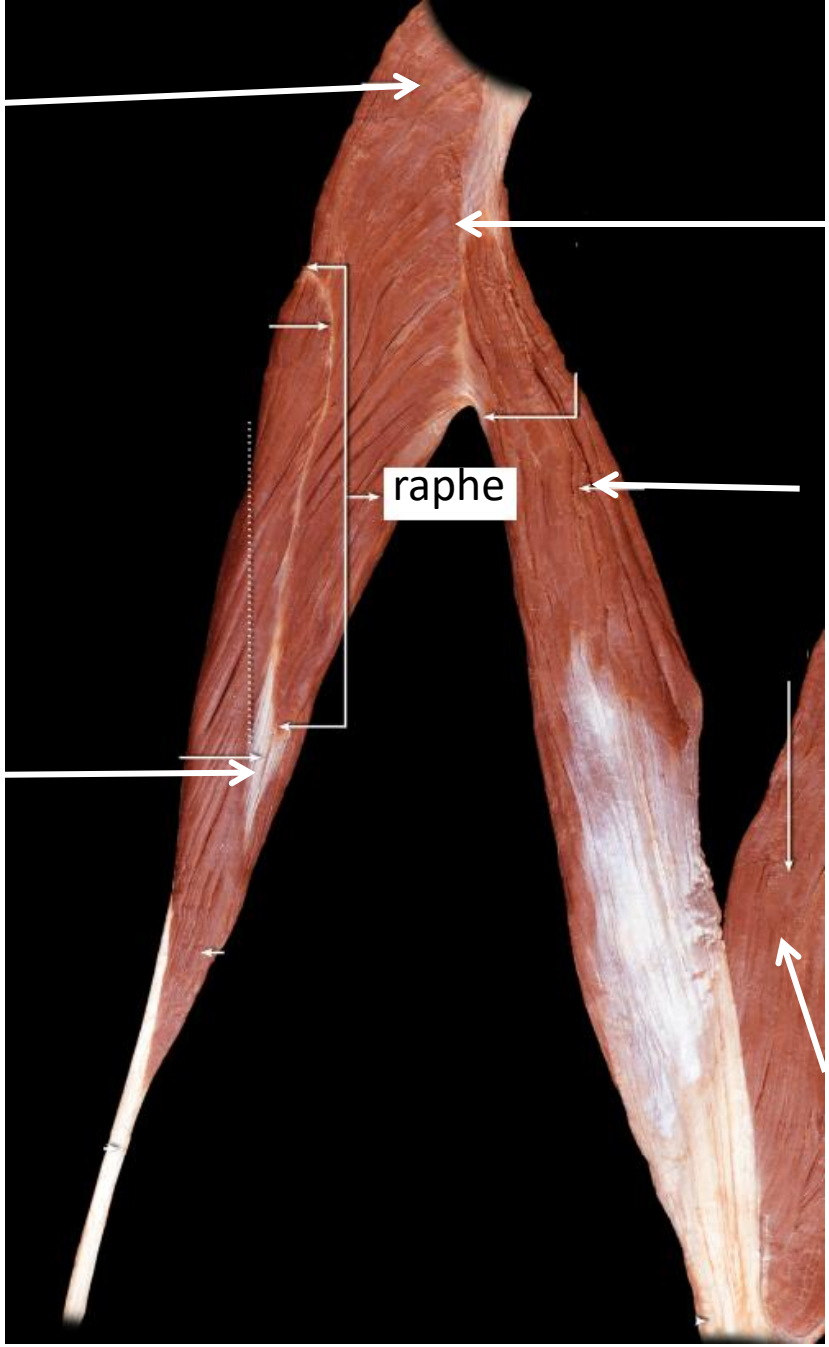


all have common atttachment at ischial tuberosity

Right Ischial Tuberosity

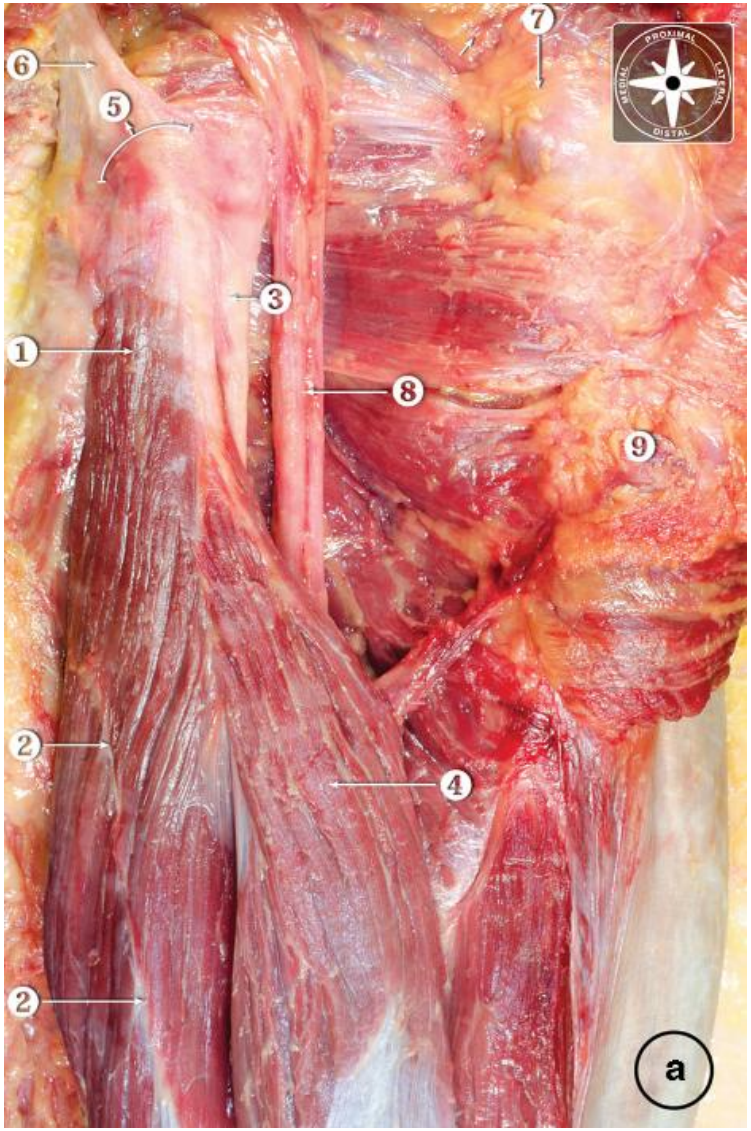


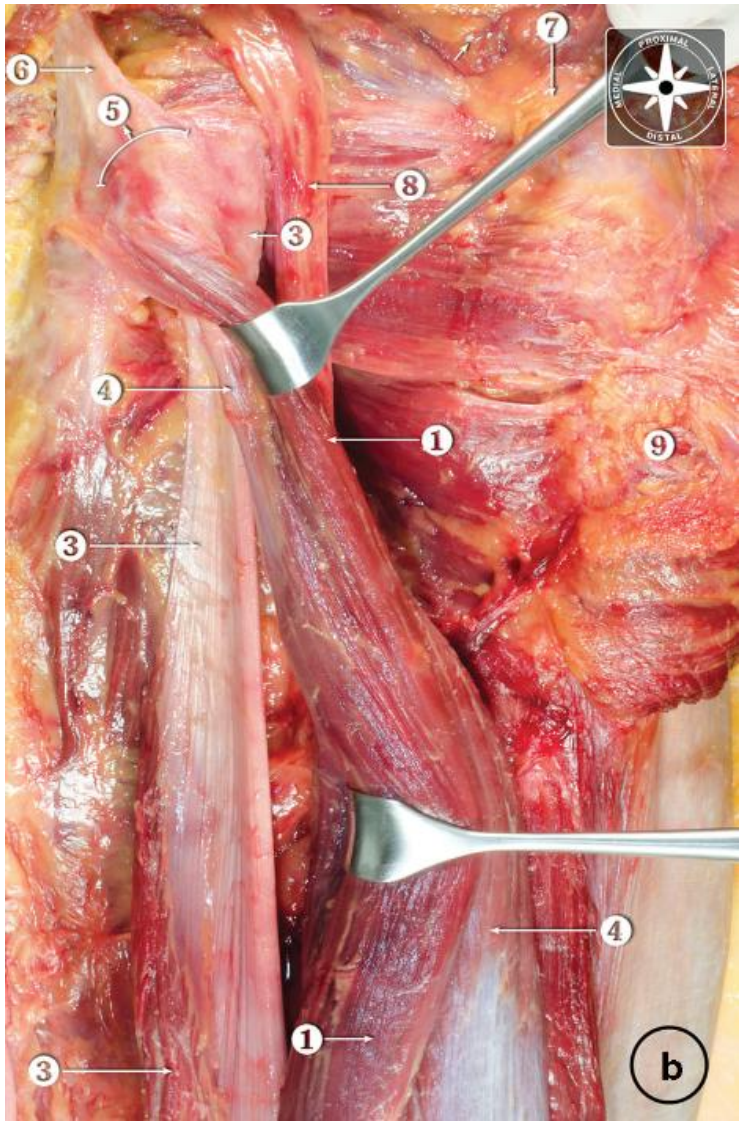

PROXIMAL
MEDIAL
LATERAL
DISTAL
Greater notch
Ischial spine
Lesser notch
Conjoint tendon
Adductor magnus



raphe- structure that connects two parts, can be seen as a white linear streak in ultrasound?xray?

Right posterior hamstring muscles

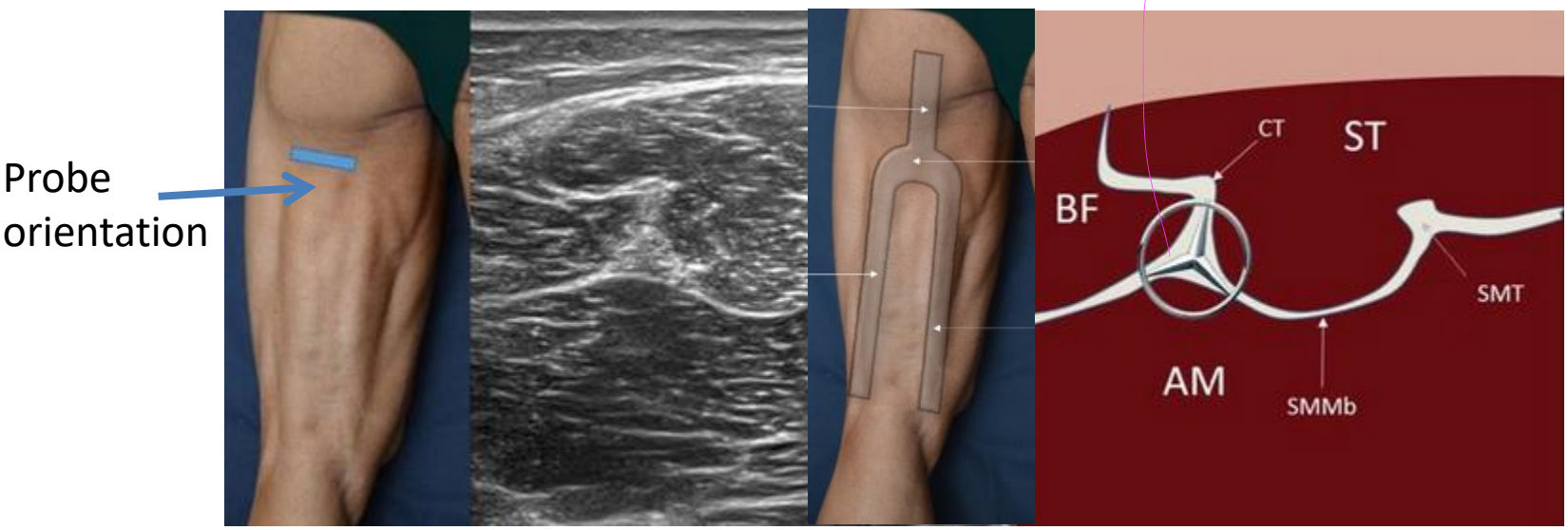




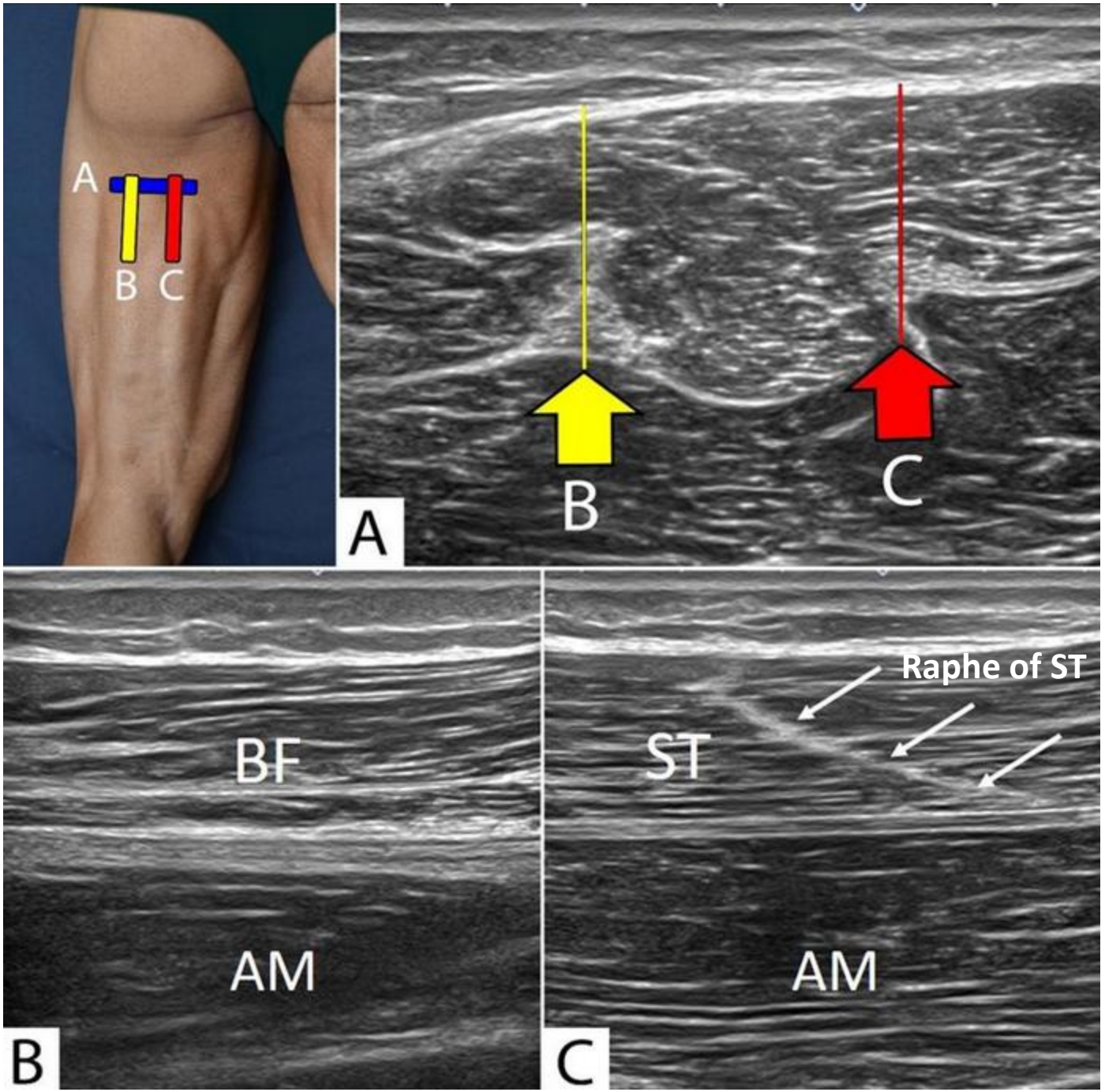
1 Semitendinosus muscle. 2 Raphe of semitendinosus muscle. 3 Semimembranosus muscle. 4 Long head of biceps femoris muscle. 5 Ischial tuberosity. 6 Sacrotuberous ligament. 7 Great trochanter. 8 Sciatic nerve 9 Gluteus maximus (cut)

Sonographic landmarks of the hamstring muscles

“Mercedes Benz sign”

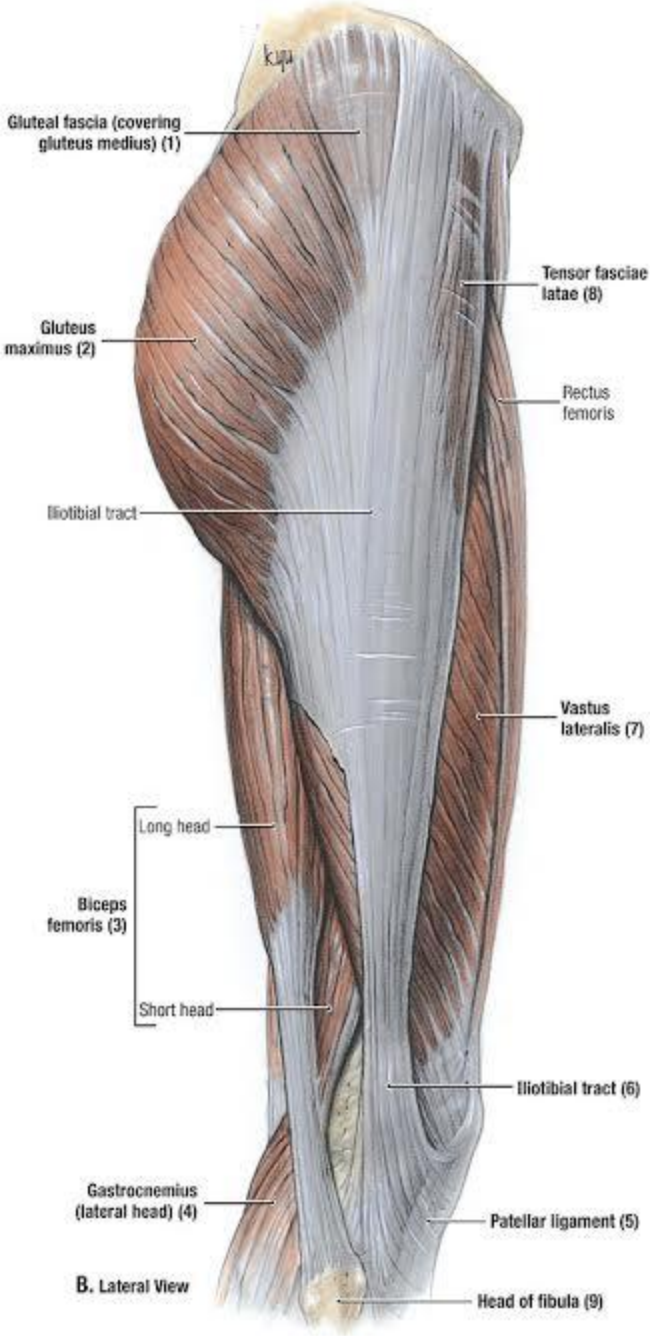


BF biceps femoris; ST semitendinosus; AM Adductor magnus; CT conjoint tendon; SMMb semimembranosus membrane ; SMT semimembranosus tendon


A
B C
B
C
A
BF
AM
B
Raphe of ST
ST
AM
C

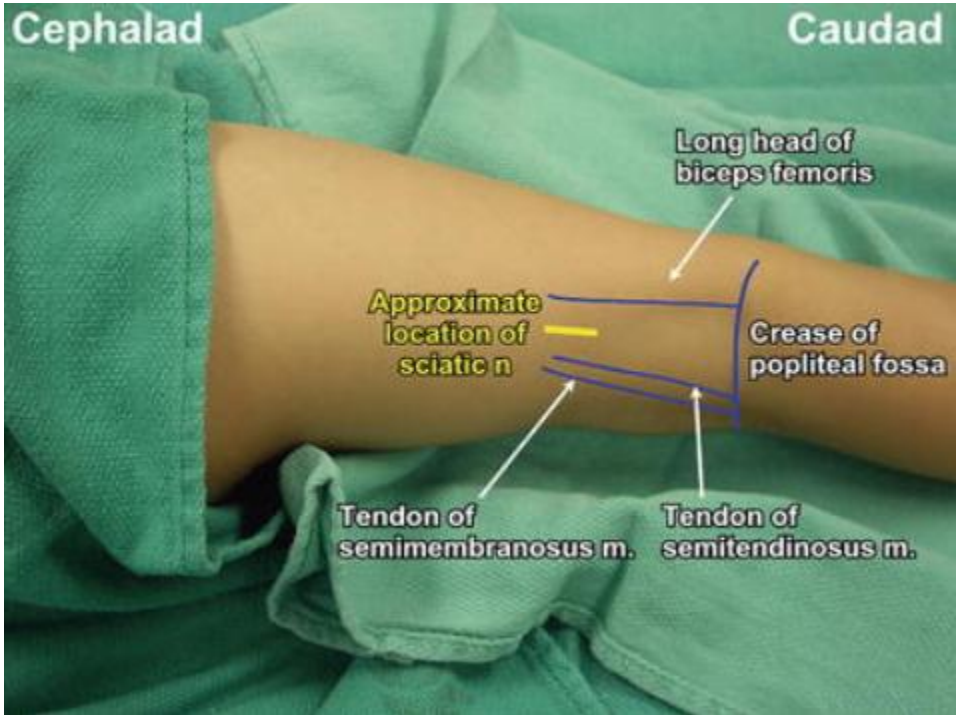


A. Lateral View



B. Lateral View

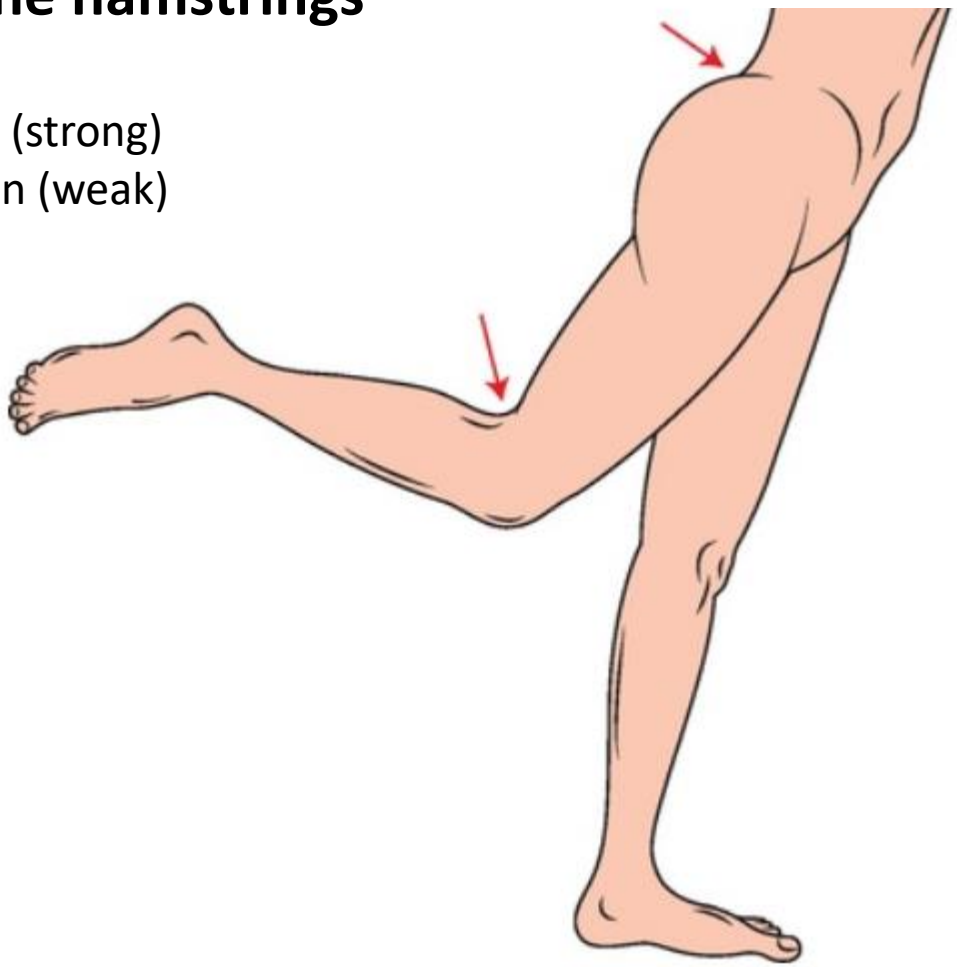
Sciatic nerve block



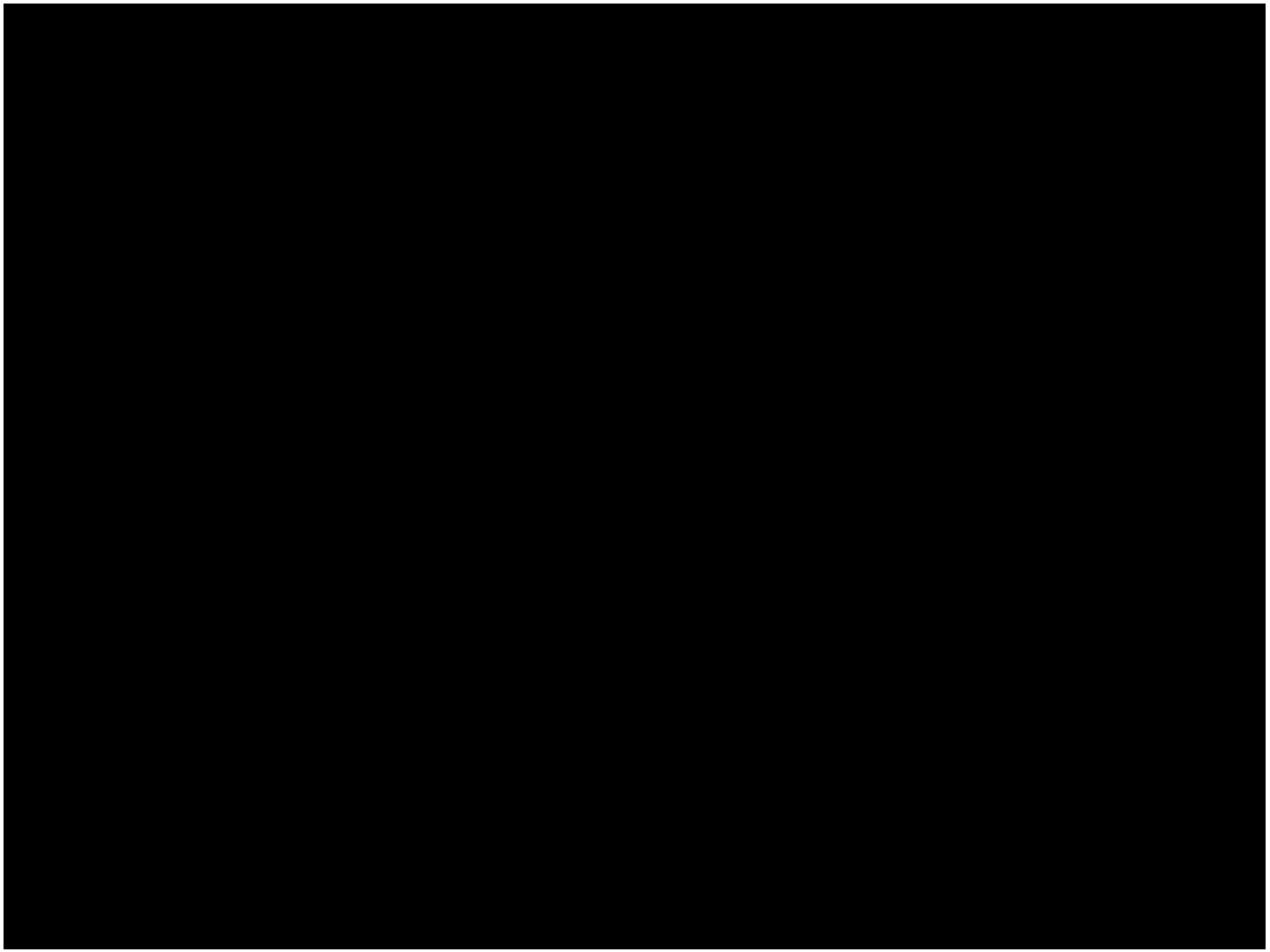
For procedures of foot and ankle

Action of the hamstrings

Knee flexion (strong)
Hip extension (weak)



Torn hamstring



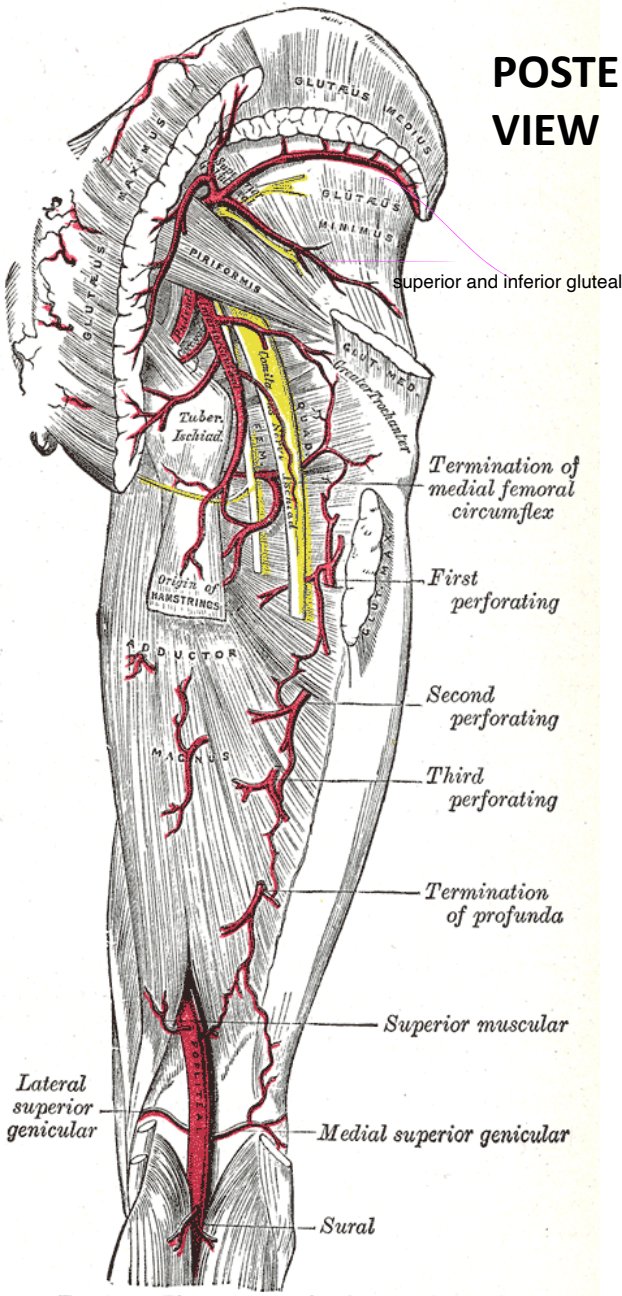
1992 Barcelona Olympics

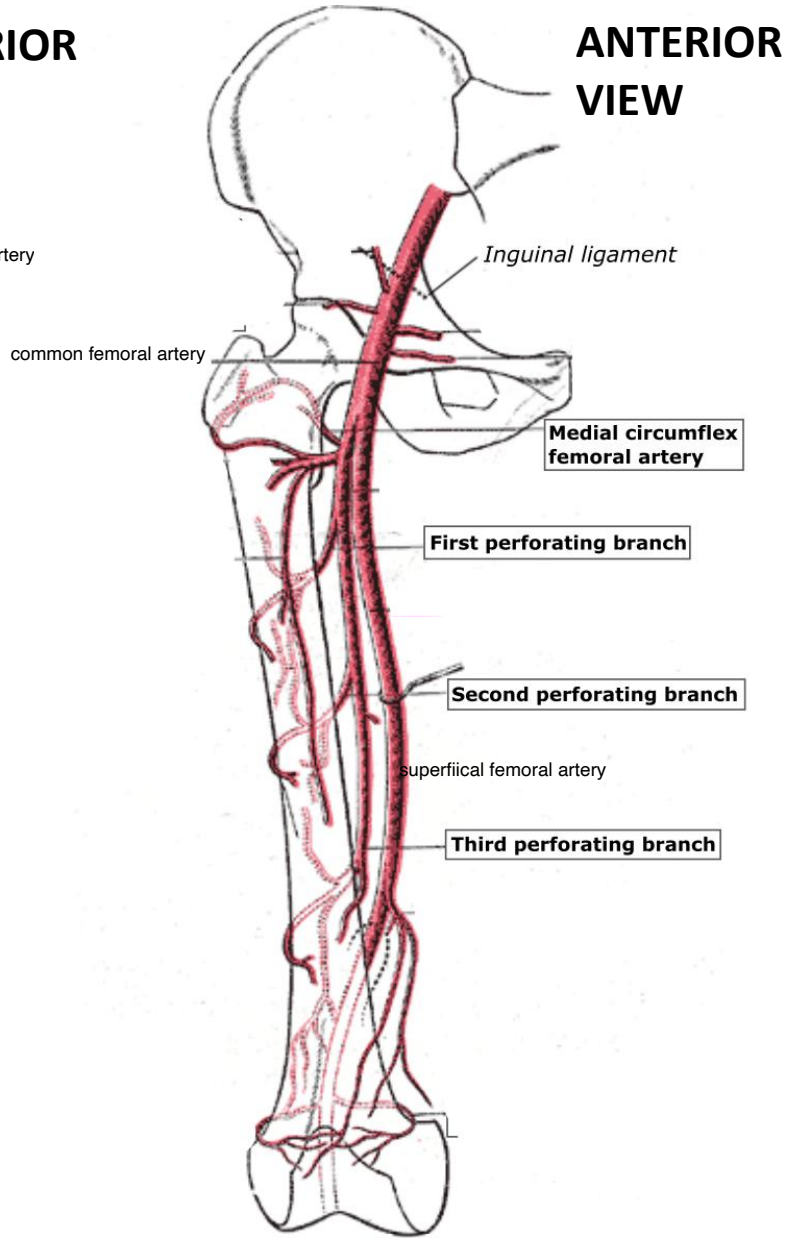
https://www.youtube.com/watch?v=t2G8KVzTwfw

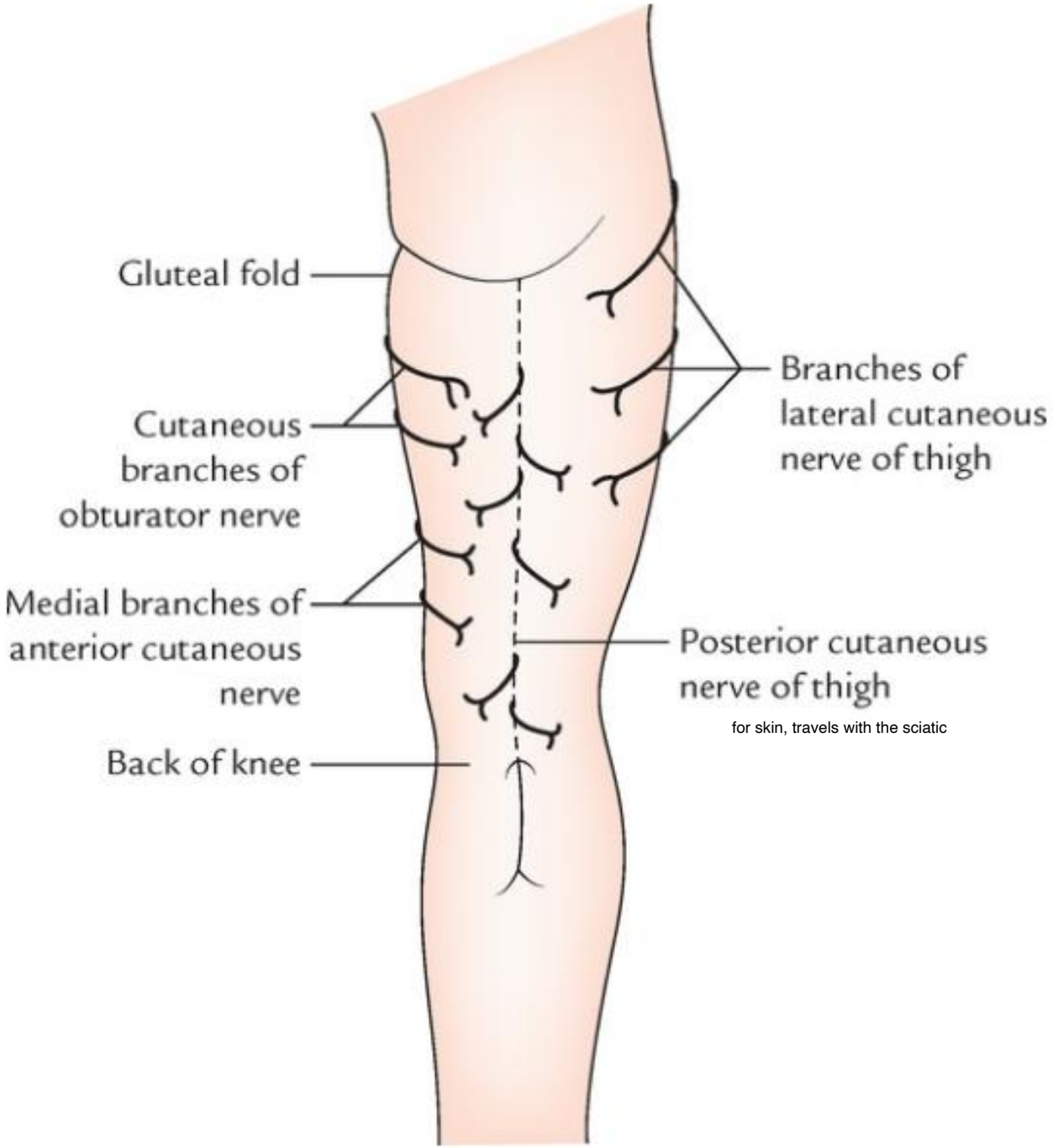
Blood supply to the posterior compartment of thigh

antalgic gait with people with torn hamstrings

The perforating branches of profunda femoris Provide a rich supply to this compartment

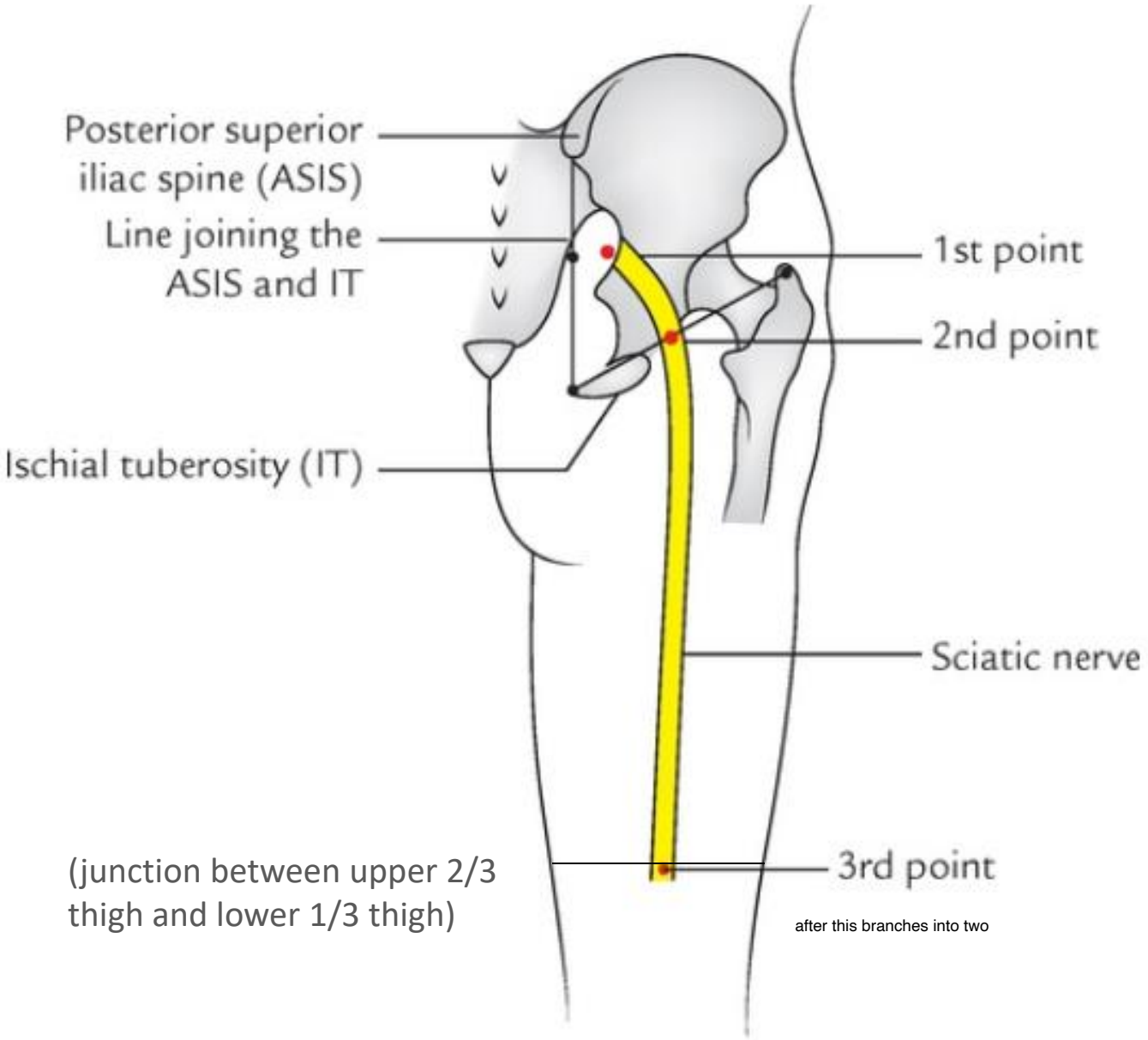






for skin, travels with the sciatic

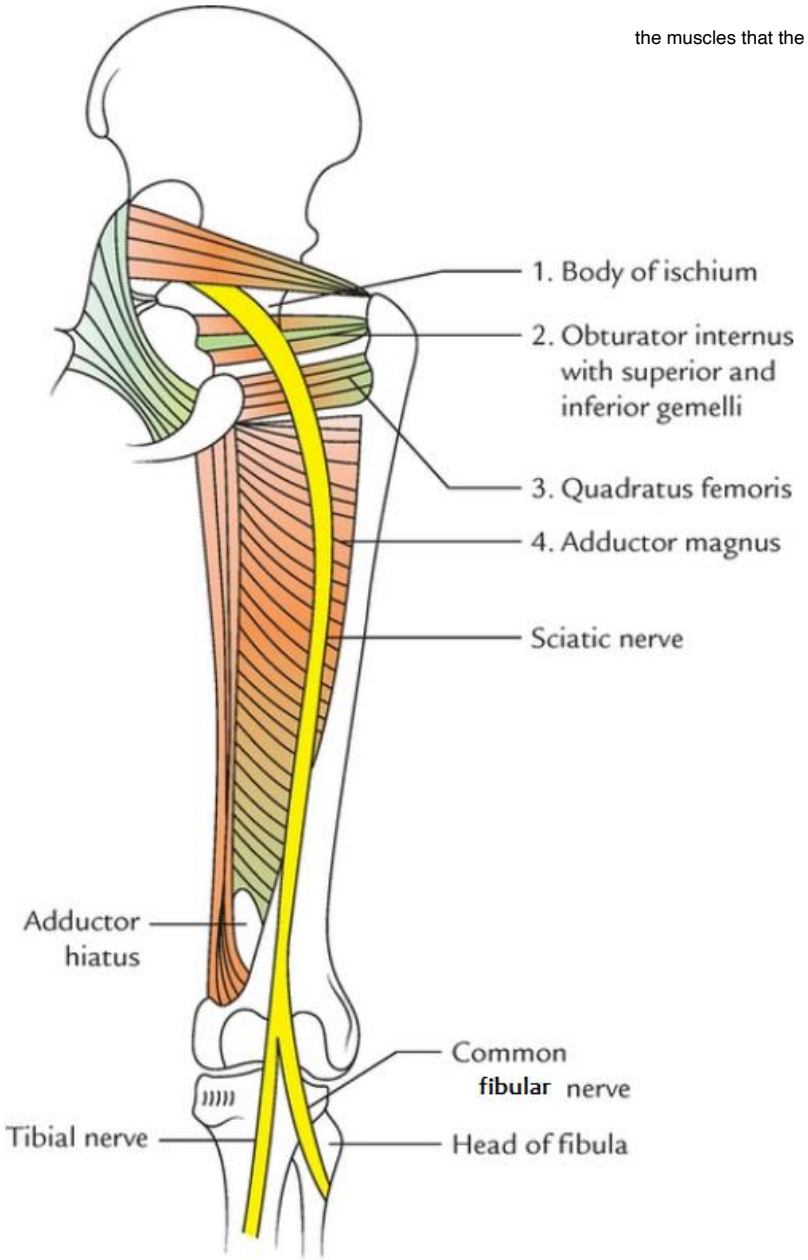
Surface markings of Sciatic nerve



(junction between upper 2/3 thigh and lower 1/3 thigh)

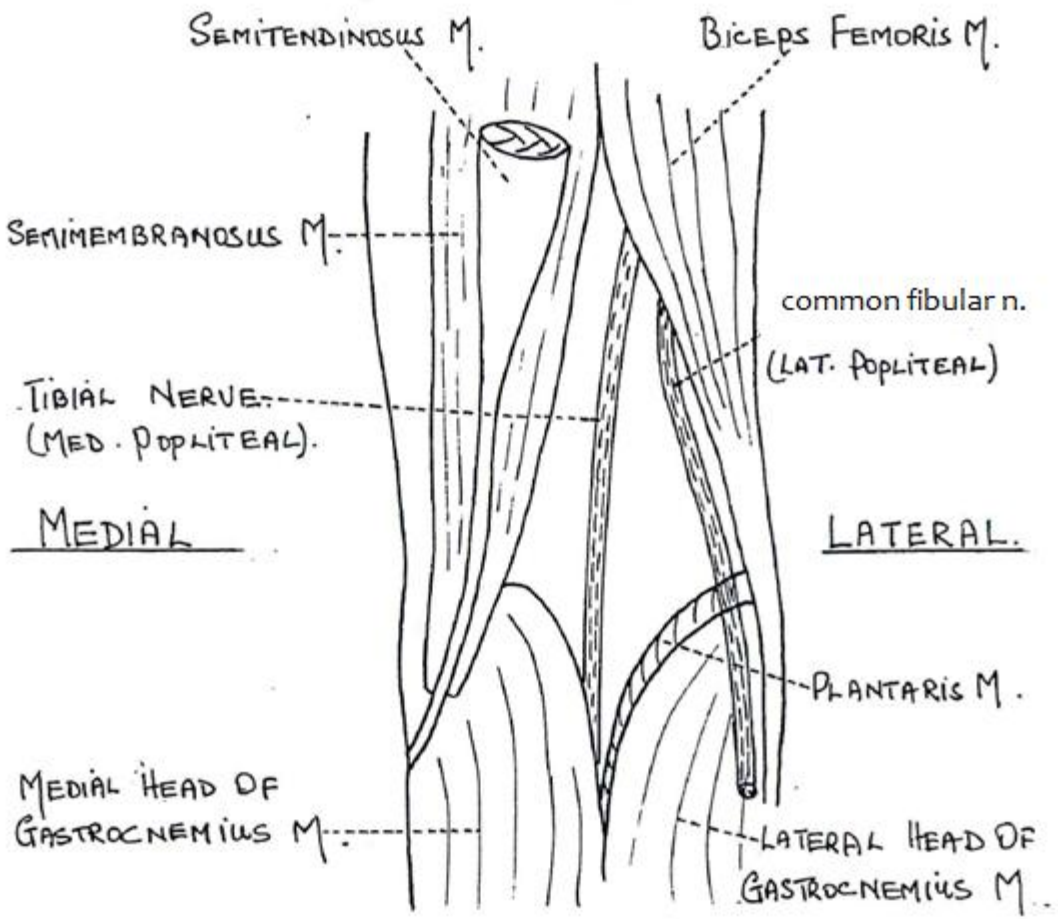
The “sciatic bed”

the muscles that the nerve "lies on"



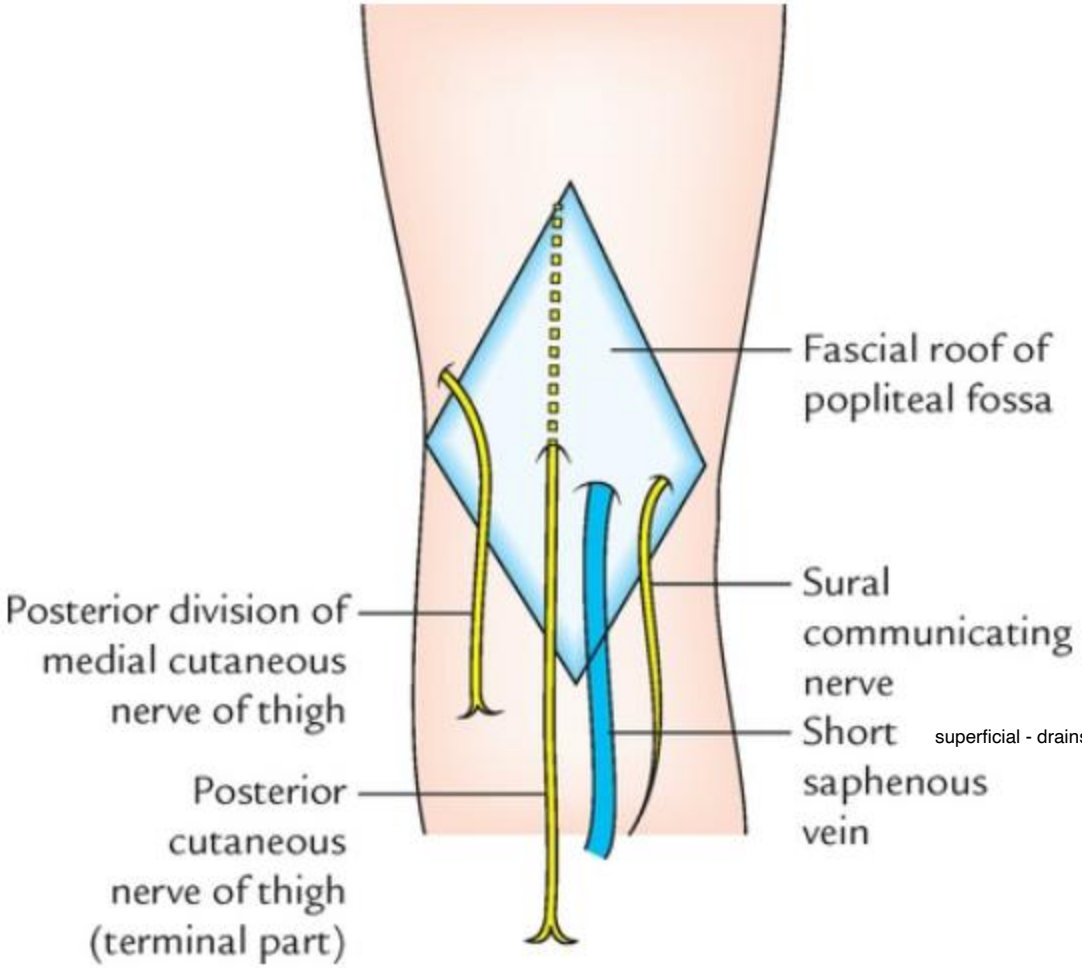
Right popliteal fossa boundaries

posterior thight



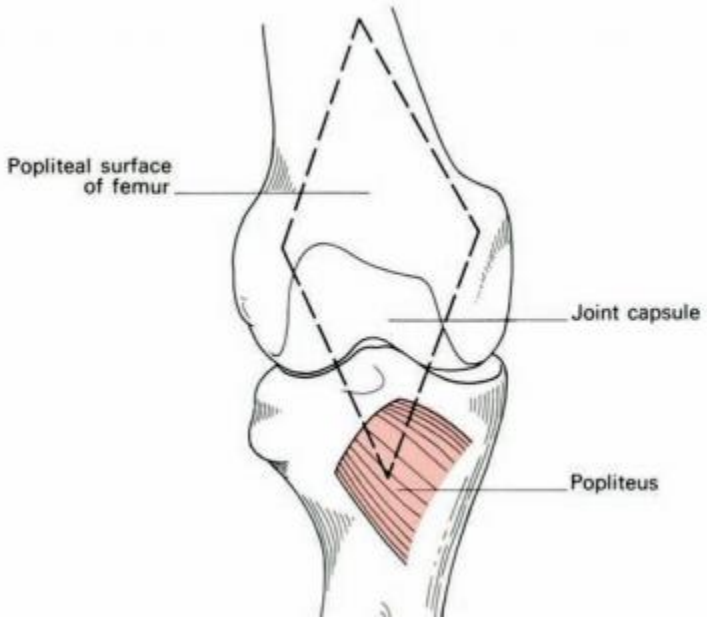
Roof of the popliteal fossa

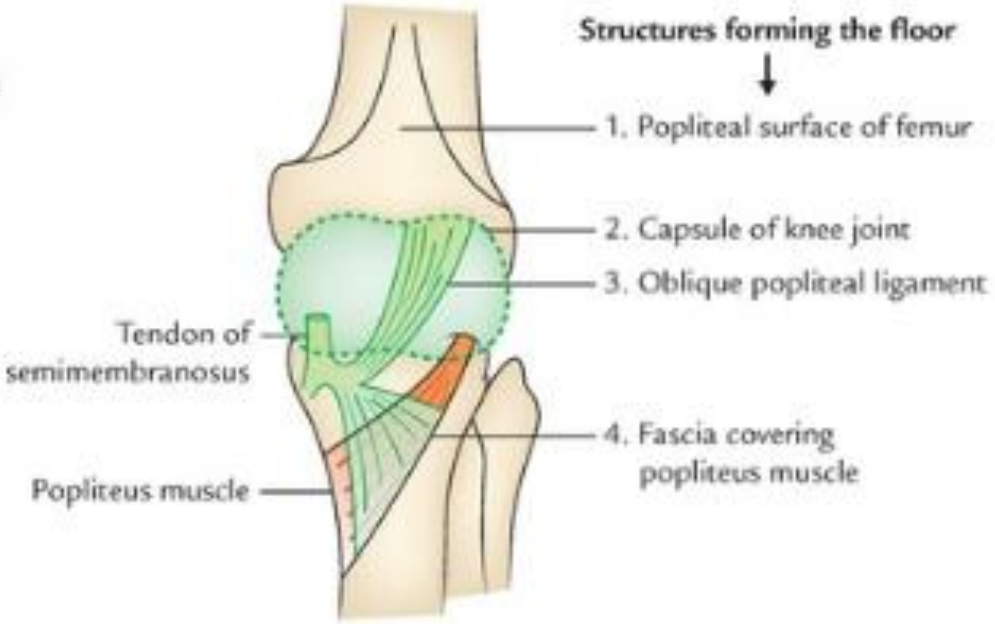
superficial aspects



superficial - drains into popliteal vein

Floor of the popliteal fossa



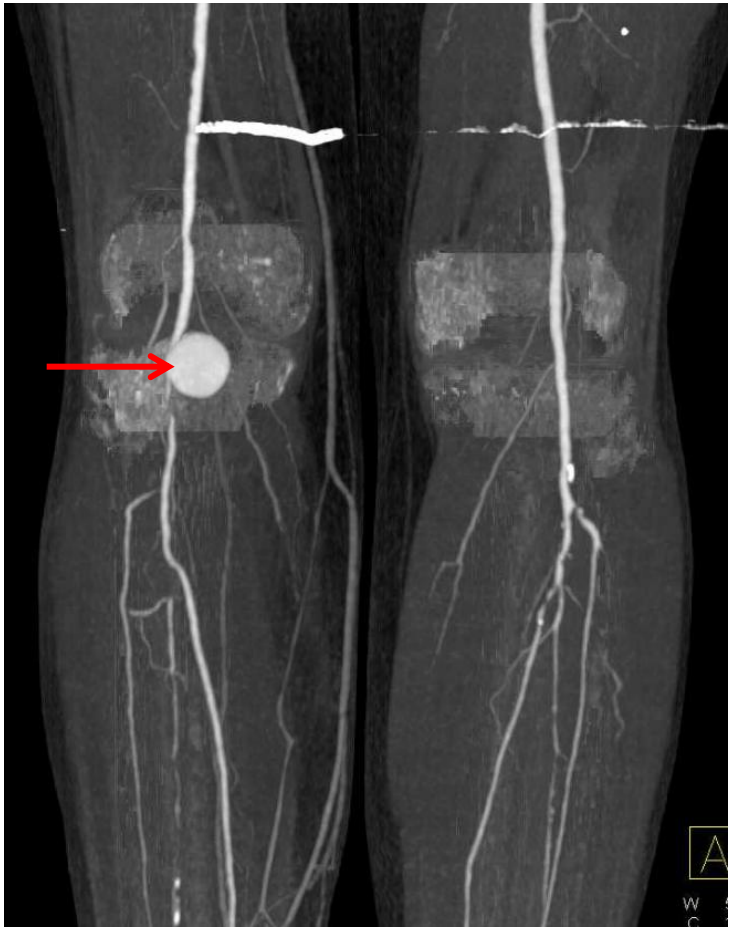


A lump in the popliteal fossa

first check if if it is cystic of solid, in ultrasound
solid is white but cystic has black streaky
lines, or shine a light if no ultrasound

skin and soft tissues—	sebaceous cyst, lipoma, sarcoma;
vein—	varicosities of the short saphenous vein in the roof of the fossa;
artery—	popliteal aneurysm;
lymph nodes—	infection secondary to suppuration in the foot;
knee joint—	joint effusion;
tendons—	enlarged bursae beneath semimembranosus/gastrocnemius;
bones—	a tumour of the lower end of femur or upper end of tibia.

Popliteal aneurysm is the most common peripheral aneurysm





- 80% of all aneurysms
- 90% occur in males
- Mean age is 65 yrs

John Hunter (1728-1793)